

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
6th Semester of B. Pharm Examination (Old Syllabus)
University Theory Examination May 2017
PH311 PHARMACEUTICAL BIOCHEMISTRY-II

Date: 12.05.2017, Friday Time: 10.00 a.m. to 1.00 p.m Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any **TWO** of the followings

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|---|--|-----------|
| A | Describe de novo synthesis of purine nucleus and conversion of IMP to AMP and GMP. | 10 |
| B | 1. Write a short note on RNA. | 05 |
| | 2. Discuss in detail about salvage pathway for purine biosynthesis. | 05 |
| C | 1. Write about the reactions involved in the heme biosynthesis. | 05 |
| | 2. Write a short note on sulphur cycle. | 05 |

Q 2 Attempt any **TWO** of the followings

- | | | |
|---|---|-----------|
| A | 1. Discuss Genetic code with its characteristics. | 05 |
| | 2. Write a note on DNA repair mechanisms. | 05 |
| B | What is replication? Why DNA replication is called semi-conservative, semi-discontinuous and bidirectional? Write a brief note about the process of replication of DNA. | 10 |
| C | 1. Outline the steps involved in the Urea cycle and state its significance. | 05 |
| | 2. What is nitrogen balance? Explain its significance and enlist the conditions causing deviations in the nitrogen balance. | 05 |

SECTION- II

Q 3 Attempt any **TWO** of the following

- A Classify kidney function tests and explain clearance tests in detail. **10**
- B Write down the chemical structure, sources and biological functions of vitamin A. **10**
- C Enumerate different types of enzymes involved in biological oxidation and explain any two with suitable reactions. **10**

Q 4 Attempt any **TWO** of the following

- A Add a detailed note on oxidative phosphorylation. **10**
- B Add a note on following: **10**
 - 1. ELISA
 - 2. PCR
- C Add a detailed note on electrophoresis. **10**